Evaluation runs: lock (a3-lock)

generated 2026-11-03 from brain-data b2ae60f86 2026-11-03 17:57:09 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): fifo: expected conforms; control-lifo: expected conforms; wrong-round: expected violation; double-grant: expected violation; double-grant: expected conforms (one client); never-grant: expected violation; never-grant: expected conforms (one client). Blue rows are at the current pins (TraceForge 9a0629e, merge join 367cdc6, JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{spec}, G_{impl} = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G^{expl}_{spec}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕒 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair control-lifo, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2,2	6	28	conforms	0.0	2	1 h, 250 GB	6	21	brain03, 2026-11-02
2,3	20	212	conforms	0.1	4	1 h, 250 GB	20	210	brain03, 2026-11-02
2,4	70	1676	conforms	0.2	2	1 h, 250 GB	75	2625	brain03, 2026-11-02
2,5	252	13604	conforms	0.9	2	1 h, 250 GB	516	65867	brain03, 2026-11-02
2,6	924	112380	conforms	9.4	4	1 h, 250 GB	2424	1192017	brain03, 2026-11-02
3,2	90	2850	conforms	0.3	2	1 h, 250 GB	98	4366	brain03, 2026-11-02
3,3	1680	324282	conforms	26.7	23	1 h, 250 GB	3214	2658596	brain03, 2026-11-02
3,4	34650		🕒 (1 h)			1 h, 250 GB			brain03, 2026-11-02
4,2	2520	651168	conforms	60.6	22	1 h, 250 GB	4833	5900122	brain03, 2026-11-02
4,3			🕒 (1 h)			1 h, 250 GB			brain03, 2026-11-02
5,2			🕒 (1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair fifo, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2,2	6	28	conforms	0.0	2	1 h, 250 GB	6	21	brain03, 2026-11-02
2,3	20	212	conforms	0.0	2	1 h, 250 GB	20	210	brain03, 2026-11-02
2,4	70	1676	conforms	0.2	2	1 h, 250 GB	77	2656	brain03, 2026-11-02
2,5	252	13604	conforms	0.9	2	1 h, 250 GB	520	66350	brain03, 2026-11-02
2,6	924	112380	conforms	9.5	2	1 h, 250 GB	2505	1201363	brain03, 2026-11-02
3,2	90	2898	conforms	0.3	2	1 h, 250 GB	129	5402	brain03, 2026-11-02
3,3	1680	330252	conforms	21.2	24	1 h, 250 GB	4560	3790449	brain03, 2026-11-02
3,4	34650		🕒 (1 h)			1 h, 250 GB			brain03, 2026-11-02
4,2	2520	675960	conforms	60.2	22	1 h, 250 GB	8437	10077551	brain03, 2026-11-02
4,3			🕒 (1 h)			1 h, 250 GB			brain03, 2026-11-02
5,2			🕒 (1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair double-grant, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2,2	6	70	64	violation	0.0	4	1 h, 250 GB	70	405	brain03, 2026-11-02
3,2	90	34650	34560	violation	31.8	38	1 h, 250 GB	34650	3114495	brain03, 2026-11-02
3,3	1680			🕒 (1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair **never-grant**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{spec}^{expl}	run
2,2	6	21	8	violation	0.0	4	1 h, 250 GB	14	69	brain03, 2026-11-02
3,2	90	1504	716	violation	2.3	4	1 h, 250 GB	806	68535	brain03, 2026-11-02
3,3	1680	83852	40576	violation	346.1	36	1 h, 250 GB	42256	69579720	brain03, 2026-11-02

stateless: pair **wrong-round**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{spec}^{expl}	run
2,2	6	28	28	violation	0.0	4	1 h, 250 GB	28	168	brain03, 2026-11-02
3,2	90	2898	2898	violation	6.2	4	1 h, 250 GB	2898	260820	brain03, 2026-11-02
3,3	1680	330252	330252	violation	2820.5	98	1 h, 250 GB	330252	554823360	brain03, 2026-11-02

Not run (stateless). planned but never scheduled (grid points beyond the ladder): wrong-round 2 3 fifo, never-grant 2 3 fifo, double-grant 2 3 fifo, never-grant 2 4 fifo, double-grant 2 4 fifo, wrong-round 2 4 fifo, wrong-round 2 5 fifo, double-grant 2 5 fifo, never-grant 2 5 fifo, wrong-round 2 6 fifo, never-grant 2 6 fifo, double-grant 2 6 fifo, never-grant 3 4 fifo, wrong-round 3 4 fifo, fifo 3 5 fifo, control-lifo 3 5 fifo, never-grant 3 5 fifo, wrong-round 3 5 fifo, wrong-round 3 6 fifo, never-grant 3 6 fifo, control-lifo 3 6 fifo, fifo 3 6 fifo, double-grant 4 2 fifo, never-grant 4 2 fifo, … (91 instances in all).

skipped: a smaller size had reached its limit: double-grant 3 4 fifo, double-grant 3 5 fifo, double-grant 3 6 fifo.

stateful

stateful: pair **control-lifo**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2,2	6	28	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,3	20	212	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,4	70	1676	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-02
2,5	252	13604	conforms	0.3	4	1 h, 250 GB	brain02, 2026-11-02
2,6	924	112380	conforms	1.3	2	1 h, 250 GB	brain02, 2026-11-02
3,2	90	2850	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-02
3,3	1680	324282	conforms	2.6	4	1 h, 250 GB	brain02, 2026-11-02
3,4	34650	40400514	conforms	782.0	188	1 h, 250 GB	brain02, 2026-11-02
3,5			(1 h)			1 h, 250 GB	brain02, 2026-11-02
4,2	2520	651168	conforms	6.0	4	1 h, 250 GB	brain02, 2026-11-02
4,3			(1 h)			1 h, 250 GB	brain02, 2026-11-02
5,2			(1 h)			1 h, 250 GB	brain02, 2026-11-02

stateful: pair **fifo**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2,2	6	28	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-02
2,3	20	212	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,4	70	1676	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-02
2,5	252	13604	conforms	0.3	2	1 h, 250 GB	brain02, 2026-11-02
2,6	924	112380	conforms	1.2	4	1 h, 250 GB	brain02, 2026-11-02
3,2	90	2898	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-02
3,3	1680	330252	conforms	2.8	4	1 h, 250 GB	brain02, 2026-11-02
3,4	34650	41133774	conforms	779.4	196	1 h, 250 GB	brain02, 2026-11-02
3,5			(1 h)			1 h, 250 GB	brain02, 2026-11-02
4,2	2520	675960	conforms	6.1	6	1 h, 250 GB	brain02, 2026-11-02
4,3			(1 h)			1 h, 250 GB	brain02, 2026-11-02
5,2			(1 h)			1 h, 250 GB	brain02, 2026-11-02

stateful: pair **double-grant**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2,2	6	70	64	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,3	20	924	904	violation	0.1	2	1 h, 250 GB	brain02, 2026-11-02
2,4	70	12870	12800	violation	0.5	28	1 h, 250 GB	brain02, 2026-11-02
2,5	252	184756	184504	violation	3.2	294	1 h, 250 GB	brain02, 2026-11-02
2,6	924	2704156	2703232	violation	42.5	4282	1 h, 250 GB	brain02, 2026-11-02
3,2	90	34650	34560	violation	0.6	48	1 h, 250 GB	brain02, 2026-11-02
3,3	1680	17153136	17151456	violation	171.9	15095	1 h, 250 GB	brain02, 2026-11-02
3,4				(1 h)			1 h, 250 GB	brain02, 2026-11-02
4,2	2520	63063000	63060480	violation	881.9	41278	1 h, 250 GB	brain02, 2026-11-02
4,3				(1 h)			1 h, 250 GB	brain02, 2026-11-02
5,2				(1 h)			1 h, 250 GB	brain02, 2026-11-02

stateful: pair **never-grant**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2,2	6	21	8	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,3	20	106	43	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,4	70	545	224	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-02
2,5	252	2864	1181	violation	0.2	4	1 h, 250 GB	brain02, 2026-11-02
2,6	924	15309	6320	violation	0.5	12	1 h, 250 GB	brain02, 2026-11-02
3,2	90	1504	716	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-02
3,3	1680	83852	40576	violation	1.0	34	1 h, 250 GB	brain02, 2026-11-02
3,4	34650	5077092	2448528	violation	83.2	1062	1 h, 250 GB	brain02, 2026-11-02
3,5				(1 h)			1 h, 250 GB	brain02, 2026-11-02
4,2	2520	257844	136092	violation	2.9	164	1 h, 250 GB	brain02, 2026-11-02
4,3				(1 h)			1 h, 250 GB	brain02, 2026-11-02
5,2	113400	83803176	47273640	violation	3204.8	54125	1 h, 250 GB	brain02, 2026-11-02

stateful: pair **wrong-round**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2,2	6	28	28	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,3	20	212	212	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-02
2,4	70	1676	1676	violation	0.2	4	1 h, 250 GB	brain02, 2026-11-02
2,5	252	13604	13604	violation	0.4	6	1 h, 250 GB	brain02, 2026-11-02
2,6	924	112380	112380	violation	1.7	14	1 h, 250 GB	brain02, 2026-11-02
3,2	90	2898	2898	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-02
3,3	1680	330252	330252	violation	2.9	116	1 h, 250 GB	brain02, 2026-11-02
3,4	34650	41133774	41133774	violation	556.5	6437	1 h, 250 GB	brain02, 2026-11-02
3,5				(1 h)			1 h, 250 GB	brain02, 2026-11-02
4,2	2520	675960	675960	violation	7.2	602	1 h, 250 GB	brain02, 2026-11-02
4,3				(1 h)			1 h, 250 GB	brain02, 2026-11-02
5,2				(1 h)			1 h, 250 GB	brain02, 2026-11-02

Not run (stateful). planned but never scheduled (grid points beyond the ladder): never-grant 3 6 fifo, wrong-round 3 6 fifo, never-grant 4 4 fifo, wrong-round 4 4 fifo, double-grant 4 4 fifo, wrong-round 4 5 fifo, never-grant 4 5 fifo, wrong-round 4 6 fifo, never-grant 4 6 fifo, wrong-round 5 3 fifo, double-grant 5 3 fifo, never-grant 5 3 fifo, never-grant 5 4 fifo, double-grant 5 4 fifo, wrong-round 5 4 fifo, never-grant 5 5 fifo, wrong-round 5 5 fifo, double-grant 5 5 fifo, wrong-round 5 6 fifo, double-grant 5 6 fifo, never-grant 5 6 fifo, never-grant 6 2 fifo, wrong-round 6 2 fifo, double-grant 6 2 fifo, … (38 instances in all).

skipped: a smaller size had reached its limit: double-grant 3 5 fifo, double-grant 3 6 fifo, control-lifo 3 6 fifo, fifo 3 6 fifo, control-lifo 4 4 fifo, double-grant 4 4 fifo, fifo 4 4 fifo, fifo 4 5 fifo, double-grant 4 5 fifo, control-lifo 4 5 fifo, double-grant 4 6 fifo, fifo 4 6 fifo, control-lifo 4 6 fifo, control-lifo 5 3 fifo, fifo 5 3 fifo, control-lifo 5 4 fifo, fifo 5 4 fifo, fifo 5 5 fifo, control-lifo 5 5 fifo, fifo 5 6 fifo, control-lifo 5 6 fifo, control-lifo 6 2 fifo, fifo 6 2 fifo, control-lifo 6 3 fifo, … (31 instances in all).

offline

offline: pair **control-lifo**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	28	conforms	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2026-11-02
3,2	90	2850	conforms	0.3	4	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain01, 2026-11-02
3,3	1680	324282	conforms	5.0	32	1 h, 250 GB	0.5	3.8	0.7	merge	brain01, 2026-11-03

offline: pair **fifo**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	28	conforms	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2026-11-02
3,2	90	2898	conforms	0.3	4	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain01, 2026-11-02
3,3	1680	330252	conforms	4.8	32	1 h, 250 GB	0.5	3.5	0.7	merge	brain01, 2026-11-03

offline: pair **double-grant**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	70	28	violation	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2026-11-02
3,2	90	34650	7044	violation	1.3	12	1 h, 250 GB	0.2	0.9	0.2	merge	brain01, 2026-11-03
3,3	1680	17153136	1823232	violation	287.9	4650	1 h, 250 GB	0.5	186.1	101.3	merge	brain01, 2026-11-03

offline: pair **never-grant**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	21	5	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
3,2	90	1504	176	violation	0.2	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
3,3	1680	83852	4076	violation	1.6	76	10 min, 16 GB	0.3	0.9	0.4	in-memory	brain01, 2026-11-02

offline: pair **wrong-round**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	6	28	6	violation	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2026-11-02
3,2	90	2898	372	violation	0.2	4	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain01, 2026-11-02
3,3	1680	330252	13584	violation	3.9	147	10 min, 16 GB	0.3	2.9	0.7	in-memory	brain01, 2026-11-02

Not run (offline). planned but never scheduled (grid points beyond the ladder): control-lifo 2 3 fifo, wrong-round 2 3 fifo, never-grant 2 3 fifo, fifo 2 3 fifo, double-grant 2 3 fifo, never-grant 2 4 fifo, double-grant 2 4 fifo, control-lifo 2 4 fifo, wrong-round 2 4 fifo, fifo 2 4 fifo, wrong-round 2 5 fifo, control-lifo 2 5 fifo, fifo 2 5 fifo, double-grant 2 5 fifo, never-grant 2 5 fifo, wrong-round 2 6 fifo, fifo 2 6 fifo, control-lifo 2 6 fifo, never-grant 2 6 fifo, double-grant 2 6 fifo, double-grant 3 4 fifo, control-lifo 3 4 fifo, never-grant 3 4 fifo, wrong-round 3 4 fifo, … (110 instances in all).

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **control-lifo**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	28	28	conforms	5.0	155	4 h, 250 GB	4.6	0.4	0.0	merge	brain02, 2026-11-03
3,2	2850	2850	conforms	9.0	466	10 min, 250 GB	8.6	0.3	0.0	merge	brain02, 2026-11-03
3,3	324282	324282	conforms	392.0	10520	10 min, 250 GB	387.2	3.8	1.1	merge	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **double-grant**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	70	70	conforms	8.0	151	10 min, 250 GB	7.8	0.2	0.0	merge	brain02, 2026-11-03
3,2	34650	34650	conforms	103.4	6508	10 min, 250 GB	102.0	0.9	0.5	merge	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **fifo**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	28	28	conforms	11.4	1810	4 h, 250 GB	11.3	0.1	0.0	merge	brain02, 2026-11-03
3,2	2898	2898	conforms	11.9	2780	10 min, 250 GB	11.4	0.3	0.2	merge	brain02, 2026-11-03
3,3	330252	330252	conforms	504.7	11648	10 min, 250 GB	499.7	3.5	1.4	merge	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **never-grant**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	21	21	conforms	4.8	168	4 h, 250 GB	4.6	0.2	0.0	merge	brain02, 2026-11-03
3,2	1504	1504	conforms	8.4	362	10 min, 250 GB	8.1	0.2	0.0	merge	brain02, 2026-11-03
3,3	83852	83852	conforms	200.3	6906	10 min, 250 GB	198.2	1.2	1.0	merge	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **wrong-round**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	28	28	conforms	4.6	171	4 h, 250 GB	4.4	0.2	0.0	merge	brain02, 2026-11-03
3,2	2898	2898	conforms	19.8	2130	10 min, 250 GB	19.4	0.4	0.0	merge	brain02, 2026-11-03
3,3	330252	330252	conforms	1021.9	12604	10 min, 250 GB	1016.9	3.8	1.2	merge	brain02, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). refused (an input missing at run time): double-grant 3 3 fifo.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **control-lifo**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	28	28	conforms	5.0	155	4 h, 250 GB	0.4	4.6	0.0	merge	brain02, 2026-11-03
3,2	2850	2850	conforms	9.0	466	10 min, 250 GB	0.3	8.6	0.0	merge	brain02, 2026-11-03
3,3	324282	324282	conforms	392.3	10520	10 min, 250 GB	3.8	387.2	1.4	merge	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **double-grant**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	70	70	conforms	8.0	151	10 min, 250 GB	0.2	7.8	0.0	merge	brain02, 2026-11-03
3,2	34650	34650	conforms	103.5	6508	10 min, 250 GB	0.9	102.0	0.5	merge	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **fifo**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	28	28	conforms	11.4	1810	4 h, 250 GB	0.1	11.3	0.0	merge	brain02, 2026-11-03
3,2	2898	2898	conforms	11.7	2780	10 min, 250 GB	0.3	11.4	0.0	merge	brain02, 2026-11-03
3,3	330252	330252	conforms	504.5	11648	10 min, 250 GB	3.5	499.7	1.3	merge	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **never-grant**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	21	21	conforms	4.8	168	4 h, 250 GB	0.2	4.6	0.0	merge	brain02, 2026-11-03
3,2	1504	1504	conforms	8.4	362	10 min, 250 GB	0.2	8.1	0.0	merge	brain02, 2026-11-03
3,3	83852	83852	conforms	200.0	6906	10 min, 250 GB	1.2	198.2	0.6	merge	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **wrong-round**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	28	28	conforms	4.6	171	4 h, 250 GB	0.2	4.4	0.0	merge	brain02, 2026-11-03
3,2	2898	2898	conforms	19.8	2130	10 min, 250 GB	0.4	19.4	0.0	merge	brain02, 2026-11-03
3,3	330252	330252	conforms	1021.8	12604	10 min, 250 GB	3.8	1016.9	1.2	merge	brain02, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). skipped: a smaller size had reached its limit: double-grant 3 3 fifo.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **control-lifo**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	28	28	conforms	3.6	164	1 h, 250 GB		brain02, 2026-11-03
3,2	2850	2850	conforms	10.3	1292	1 h, 250 GB		brain02, 2026-11-03
3,3	324282	324282	conforms	387.5	14271	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **double-grant**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	70	70	conforms	5.6	175	1 h, 250 GB		brain02, 2026-11-03
3,2	34650	34650	conforms	61.5	7169	1 h, 250 GB		brain02, 2026-11-03
3,3	17153136		(1 h)			1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **fifo**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	28	28	conforms	3.6	155	1 h, 250 GB		brain02, 2026-11-03
3,2	2898	2898	conforms	12.3	2959	1 h, 250 GB		brain02, 2026-11-03
3,3	330252	330252	conforms	485.6	13798	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **never-grant**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	21	21	conforms	3.6	158	1 h, 250 GB		brain02, 2026-11-03
3,2	1504	1504	conforms	10.7	2646	1 h, 250 GB		brain02, 2026-11-03
3,3	83852	83852	conforms	146.7	7630	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **wrong-round**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	28	28	conforms	3.5	166	1 h, 250 GB		brain02, 2026-11-03
3,2	2898	2898	conforms	10.5	3184	1 h, 250 GB		brain02, 2026-11-03
3,3	330252	330252	conforms	395.1	13563	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **control-lifo**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	28	28	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
3,2	2850	2850	conforms	0.2	6	1 h, 250 GB		brain02, 2026-11-03
3,3	324282	324282	conforms	4.7	136	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **double-grant**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	70	70	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
3,2	34650	34650	conforms	1.1	48	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **fifo**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	28	28	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
3,2	2898	2898	conforms	0.2	6	1 h, 250 GB		brain02, 2026-11-03
3,3	330252	330252	conforms	4.9	143	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **never-grant**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	21	21	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
3,2	1504	1504	conforms	0.1	6	1 h, 250 GB		brain02, 2026-11-03
3,3	83852	83852	conforms	1.6	70	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair wrong-round, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2	28	28	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
3,2	2898	2898	conforms	0.2	6	1 h, 250 GB		brain02, 2026-11-03
3,3	330252	330252	conforms	6.7	139	1 h, 250 GB		brain02, 2026-11-03

Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export).
skipped: a smaller size had reached its limit: double-grant 3 3 fifo.